Solution Requirements

Date	21 October 2026
Team ID	VAI-AI-2026
Project Name	VoyageAI — Personalized AI Travel Itinerary Planner
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Category	Requirement Description	Priority
1	Authentication	Secure user signup/login via Supabase Auth with asymmetric ES256 JWT claim verification and state persistence.	High
2	Authorization levels	PostgreSQL Row-Level Security (RLS) guaranteeing users only read and mutate itineraries where auth.uid() = user_id.	High
3	External interfaces	REST communication with Groq Cloud LPU (openai/gpt-oss-120b), Supabase DB pooler on Port 6543, Render, and Vercel CDN.	High
4	Transactions processing	Asynchronous SQLAlchemy 2.0 transactions saving trips, days, and activities with cascade-delete integrity.	High
5	Reporting	User dashboard tracking total trips, durations, and estimated expenditures; clean Print-to-PDF export module.	High
6	Business rules, etc.	Cumulative activity costs must not exceed total budget; activity count strictly follows selected pace (2-3, 3-4, 5-6/day).	High
7	Compliance to laws	GDPR-aligned data privacy with user data isolation, encrypted HTTPS/TLS 1.3 endpoints, and zero password logging.	Medium
8	Other (Re-roll)	Single-day AI regeneration button that re-rolls one day's activities without impacting the rest of the itinerary.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Sub-second LLM inference generation and rapid UI rendering.	≤ 1.5s AI response time; page loads < 800ms on Vercel CDN.
2	Scalability	Non-blocking async event loop handling concurrent user volume.	Supports 500+ concurrent requests via asyncpg pooler (port 6543).
3	Security & Privacy	End-to-end data encryption and complete multi-tenant isolation.	100% database queries guarded by RLS; zero secret key leaks on client.
4	Reliability	High cloud uptime with fault-tolerant JSON self-healing fallback parser.	99.9% uptime on Render/Vercel; zero malformed JSON crashes.
5	Usability	Mobile-first responsive UI with intuitive visual hierarchy and icons.	WCAG 2.1 AA accessible contrast; responsive on mobile, tablet, and PC.
6	Maintainability	Clean monorepo separation of frontend, backend, and schemas.	Decoupled REST interfaces with strict Pydantic v2 schemas.